Week 02 Study Guide

Linear Regression

Foundations of Artificial Intelligence - SEMTM0051

Based on Lecture 2 slides by Dr Roberto Colomboni

What the lecture is about. The lecture explains how to predict a numerical target from inputs, measure prediction error with a loss function, choose parameters by least-squares fitting, and improve parameters iteratively with gradient descent. It then extends the same ideas to nonlinear features and several input variables.

1. The Core Regression Problem

We observe N input-target pairs:

$$(x_1, y_1), \ldots, (x_N, y_N) \in \mathbb{R}^2.$$

The lecture models each observed target as

$$\boxed{y_i = f_{\text{true}}(x_i) + \varepsilon_i}, \qquad i = 1, \ldots, N.$$

Symbol	Meaning
N	Number of observations in the dataset.
i	Observation index: $i = 1, \ldots, N$.
x_i	Input value of observation i.
y_i	Observed target value of observation i.
f_{true}	The true underlying relationship between input and target. It is unknown.
ε_i	Noise term for observation i. It is also unknown.

Affine relationship

A first special case is an affine relationship:

$$\boxed{f_{\text{true}}(x) = ax + b}, \qquad y_i = ax_i + b + \varepsilon_i.$$

Here a is the **slope** and b is the **intercept**, also called the **bias**.

Important distinction: During fitting, the observed x_i and y_i are fixed. The model parameters are the quantities we vary. The slides state this directly: *the data stay fixed; the parameters vary.*

2. Variables, Parameters, Predictions, and Residuals

Input variable vs. model parameters

For an affine predictor,

$$\boxed{f_{\boldsymbol{\theta}}(x) = b + ax}.$$

Quantity	Type	Meaning
x	input variable	A value supplied to the predictor.
x_i	observed data	Input in training observation i.
y_i	observed data	Target we want the model to predict for observation i.
a	parameter	Adjustable slope.
b	parameter	Adjustable intercept/bias.
$\boldsymbol{\theta}$	parameter vector	Collects all adjustable parameters.

The parameters are collected into a column vector:

$$\boxed{\boldsymbol{\theta} = \begin{bmatrix} b \\ a \end{bmatrix}}.$$

The symbol $\boldsymbol{\theta}$ therefore means **the collection of all adjustable coefficients of the predictor**.

Prediction

For an observed input x_i, the model predicts

$$\boxed{\hat{y}_i = f_{\boldsymbol{\theta}}(x_i) = b + ax_i}.$$

The "hat" in $\hat{y}_i$ means **predicted value**; y_i is the observed target.

Residual

The slides define the residual as

$$\boxed{r_i = \text{prediction} - \text{target} = f_{\boldsymbol{\theta}}(x_i) - y_i}.$$

For the affine model,

$$\boxed{r_i(b, a) = b + ax_i - y_i}.$$

A positive residual means the prediction is above the target; a negative residual means it is below the target.

Mental model: x_i, y_i describe the dataset. a, b describe the model. The model uses x_i to produce $\hat{y}_i$, and the residual compares $\hat{y}_i$ with y_i.

3. Loss Functions and Empirical Loss

A loss function assigns a numerical penalty to a prediction $\hat{y}$ and target y.

Absolute loss

$$\boxed{\ell_{\text{abs}}(\hat{y}, y) = |\hat{y} - y|}$$

If the residual doubles, the absolute penalty doubles.

Squared loss

$$\boxed{\ell_{\text{sq}}(\hat{y}, y) = (\hat{y} - y)^2}$$

If the residual doubles, the squared penalty becomes four times larger.

Squared loss is the focus of the lecture because it is differentiable everywhere.

One-observation loss vs. total empirical loss

$$\ell_{\text{sq}}(f_{\boldsymbol{\theta}}(x_i), y_i) = \big(f_{\boldsymbol{\theta}}(x_i) - y_i\big)^2$$

is the loss for **one** observation.

Adding losses over all observations gives the empirical squared loss:

$$\boxed{\mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = \sum_{i=1}^{N} \big(f_{\boldsymbol{\theta}}(x_i) - y_i\big)^2 = \sum_{i=1}^{N} r_i^2.}$$

Notation to remember: ℓ is the loss for one prediction; $\mathcal{L}$ is the loss over the entire observed dataset.

The fitting objective

Training means choosing the parameter values that minimise empirical loss:

$$\boxed{\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \mathcal{L}_{\text{sq}}(\boldsymbol{\theta}).}$$

The star * denotes an optimal value.

At a differentiable local minimum, Fermat's theorem gives

$$\boxed{\nabla \mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = \mathbf{0}.}$$

Therefore the lecture first finds stationary points and then checks whether they are minima.

4. Least-Squares Fitting: Three Model Families

A. Constant predictor

Set $a = 0$. The predictor always returns the same value:

$$f_b^{\text{const}}(x) = b, \qquad \mathcal{L}_{\text{sq}}^{\text{const}}(b) = \sum_{i=1}^{N} (b - y_i)^2.$$

Differentiate and set the derivative to zero:

$$\frac{d\mathcal{L}}{db} = 2 \sum_{i=1}^{N} (b - y_i) = 0.$$

Hence

$$\boxed{b^* = \frac{1}{N} \sum_{i=1}^{N} y_i = \bar{y}.}$$

So the best constant predictor under squared loss is the **sample mean of the targets**.

B. Line through the origin

Set $b = 0$. Only the slope varies:

$$f_a^{\text{slope}}(x) = ax, \qquad \mathcal{L}_{\text{sq}}^{\text{slope}}(a) = \sum_{i=1}^{N} (ax_i - y_i)^2.$$

The best slope is

$$\boxed{a^* = \frac{\sum_{i=1}^{N} x_i y_i}{\sum_{i=1}^{N} x_i^2}}, \qquad \text{if } \sum_{i=1}^{N} x_i^2 > 0.$$

If every $x_i = 0$, every slope gives the same predictions and loss.

C. Full affine predictor

Both parameters vary:

$$\boxed{f_{\boldsymbol{\theta}}(x) = b + ax}, \qquad \boxed{\mathcal{L}_{\text{sq}}^{\text{aff}}(b, a) = \sum_{i=1}^{N} (b + ax_i - y_i)^2.}$$

Now the objective is a function of **two parameters**, so we use two partial derivatives.

5. Full Affine Fit: Derivatives and Closed-Form Solution

Define the residual

$$r_i(b, a) = b + ax_i - y_i.$$

Derivative with respect to the intercept

When differentiating with respect to b, hold a, x_i, y_i fixed:

$$\boxed{\frac{\partial \mathcal{L}}{\partial b} = 2\sum_{i=1}^{N}(b + ax_i - y_i) = 2\left(Nb + a\sum_{i=1}^{N} x_i - \sum_{i=1}^{N} y_i\right).}$$

Derivative with respect to the slope

When differentiating with respect to a, hold b, x_i, y_i fixed:

$$\boxed{\frac{\partial \mathcal{L}}{\partial a} = 2\sum_{i=1}^{N} x_i(b + ax_i - y_i) = 2\left(b\sum_{i=1}^{N} x_i + a\sum_{i=1}^{N} x_i^2 - \sum_{i=1}^{N} x_i y_i\right).}$$

Set both derivatives to zero. This gives a linear system in the unknowns b and a.

Define the empirical means:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad \bar{y} = \frac{1}{N}\sum_{i=1}^{N} y_i.$$

If the inputs are not all equal, the solution is

$$\boxed{a^* = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N}(x_i - \bar{x})^2}}$$

and

$$\boxed{b^* = \bar{y} - a^*\bar{x}.}$$

Lecture dataset

For the observations used in the slides,

$$(x_i) = (-2, -1, 0, 1, 2), \qquad (y_i) = (-1.5, 0.5, 1, 2.5, 7.5).$$

The empirical means are $\bar{x} = 0$ and $\bar{y} = 2$, giving

$$\boxed{a^* = 2, \qquad b^* = 2, \qquad f_{\boldsymbol{\theta}^*}(x) = 2 + 2x.}$$

The squared empirical loss at this fit is

$$\boxed{\mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}^*) = 6.}$$

6. Gradient and Gradient Descent

For the affine model, collect the partial derivatives into the gradient:

$$\boxed{\nabla \mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}) = \begin{bmatrix} \dfrac{\partial \mathcal{L}}{\partial b} \\ \dfrac{\partial \mathcal{L}}{\partial a} \end{bmatrix}.}$$

The gradient points in the direction of increasing loss locally, so a small step in the **opposite** direction decreases the loss when the gradient is nonzero.

Gradient-descent update

$$\boxed{\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta \nabla \mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}^{(t)}).}$$

Symbol	Meaning
t	Iteration number.
$\boldsymbol{\theta}^{(t)}$	Current parameter vector at iteration t.
$\eta > 0$	Learning rate / step size.
$\nabla \mathcal{L}(\boldsymbol{\theta}^{(t)})$	Gradient evaluated at the current parameters.

First gradient-descent step in the slides

The lecture starts from

$$\boldsymbol{\theta}^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \eta = 0.04.$$

At this point,

$$\frac{\partial \mathcal{L}}{\partial b}(0,0) = -20, \qquad \frac{\partial \mathcal{L}}{\partial a}(0,0) = -40.$$

Therefore

$$\boldsymbol{\theta}^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.04 \begin{bmatrix} -20 \\ -40 \end{bmatrix} = \boxed{\begin{bmatrix} 0.8 \\ 1.6 \end{bmatrix}}.$$

The new predictor is

$$\boxed{f_{\boldsymbol{\theta}^{(1)}}(x) = 0.8 + 1.6x.}$$

The slides show successive iterations moving toward the optimum near $(b, a) = (2, 2)$.

Why this matters: For this simple model we can solve for the optimum explicitly. More complex models may not have a convenient explicit solution, but gradients can still be used to improve their parameters step by step.

7. Extensions: Nonlinear Features and Several Inputs

Nonlinear predictor in the input

The lecture extends the affine predictor to, for example,

$$\boxed{f_{\boldsymbol{\theta}}(x) = b + w_1 x + w_2 x^2}, \qquad \boldsymbol{\theta} = \begin{bmatrix} b \\ w_1 \\ w_2 \end{bmatrix}.$$

This predictor is nonlinear in x, but remains linear in the adjustable coefficients.

Define a feature row vector

$$\boxed{\boldsymbol{\phi}(x) = \begin{bmatrix} 1 & x & x^2 \end{bmatrix}}.$$

Then

$$\boxed{f_{\boldsymbol{\theta}}(x) = \boldsymbol{\phi}(x)\boldsymbol{\theta}.}$$

More generally, choose fixed feature functions $\phi_1, \ldots, \phi_m$ and define

$$\boldsymbol{\phi}(x) = \begin{bmatrix} \phi_1(x) & \cdots & \phi_m(x) \end{bmatrix},$$

$$\boxed{f_{\boldsymbol{\theta}}(x) = \boldsymbol{\phi}(x)\boldsymbol{\theta} = \sum_{j=1}^{m} \theta_j \phi_j(x).}$$

Here m is the number of features and j indexes those features.

Feature matrix

For all observations, define $\Phi_{ij} = \phi_j(x_i)$. Then

$$\boldsymbol{\Phi} \in \mathbb{R}^{N \times m}, \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^{N \times 1}.$$

The squared loss becomes

$$\boxed{\mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = \|\boldsymbol{\Phi}\boldsymbol{\theta} - \mathbf{y}\|_2^2.}$$

If $\boldsymbol{\Phi}^\top \boldsymbol{\Phi}$ is invertible,

$$\boxed{\boldsymbol{\theta}^* = (\boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^\top \mathbf{y}.}$$

8. Several Input Variables and the Design Matrix

An observation may contain d input features:

$$\mathbf{x} = \begin{bmatrix} x_1 & \cdots & x_d \end{bmatrix} \in \mathbb{R}^{1\times d}.$$

The lecture's example is predicting salary from features such as age, education, and years of experience.

Associate one weight with each feature:

$$\mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_d \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} b \\ \mathbf{w} \end{bmatrix}.$$

Then

$$\boxed{f_{\boldsymbol{\theta}}(\mathbf{x}) = b + \mathbf{x}\mathbf{w} = b + \sum_{j=1}^{d} x_j w_j.}$$

Design matrix

Stack all N observations into

$$\boxed{\mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1d} \\ \vdots & \vdots & & \vdots \\ 1 & x_{N1} & \cdots & x_{Nd} \end{bmatrix} \in \mathbb{R}^{N\times(d+1)}.}$$

- Each **row** represents one observation.
- Each input-feature **column** represents one feature.
- The first column contains ones and accounts for the bias b.

With

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix},$$

the squared loss is

$$\boxed{\mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2.}$$

If $\mathbf{X}^\top\mathbf{X}$ is invertible,

$$\boxed{\boldsymbol{\theta}^* = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}.}$$

The gradient is

$$\boxed{\nabla\mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = 2\mathbf{X}^\top(\mathbf{X}\boldsymbol{\theta} - \mathbf{y}).}$$

Gradient descent therefore becomes

$$\boxed{\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - 2\eta\,\mathbf{X}^\top(\mathbf{X}\boldsymbol{\theta}^{(t)} - \mathbf{y}).}$$

9. One-Page Concept Map

1. Observe data
$(x_i, y_i),\ i = 1, \ldots, N$

$\Downarrow$

2. Choose a predictor
$f_{\boldsymbol{\theta}}(x)$, with adjustable parameters $\boldsymbol{\theta}$

$\Downarrow$

3. Predict
$\hat{y}_i = f_{\boldsymbol{\theta}}(x_i)$

$\Downarrow$

4. Compute residuals
$r_i = \hat{y}_i - y_i$

$\Downarrow$

5. Compute total loss
$\mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = \sum_i r_i^2$

$\Downarrow$

6. Find better parameters
Closed form when available, or gradient descent:
$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta \nabla \mathcal{L}(\boldsymbol{\theta}^{(t)})$

Central idea of the lecture: Linear regression learns model parameters from observed data by choosing the parameter values that minimise prediction error under a chosen loss function.

Notation checklist

Symbol	Meaning	Symbol	Meaning
N	observations	i	observation index
x_i	input data	y_i	target data
$\hat{y}_i$	prediction	r_i	residual
a	slope	b	intercept / bias
$\boldsymbol{\theta}$	parameter vector	$\boldsymbol{\theta}^*$	optimal parameters
ℓ	one-example loss	$\mathcal{L}$	empirical loss
η	learning rate	t	iteration number
d	input features	m	feature functions
$\mathbf{X}$	design matrix	$\boldsymbol{\Phi}$	feature matrix

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